

OBSERVATION TECHNIQUES, ADVANTAGES, DISADVANTAGES AND PERFORMANCE ANALYSIS OF PCA-BASED ENERGY CONSUMPTION PREDICTION MODELS

INTRODUCTION

Energy consumption forecasting is an important part of smart energy management systems. Accurate prediction helps reduce energy wastage, improve power distribution, optimize operational efficiency, and lower greenhouse gas emissions. Traditional prediction models often struggle with large datasets because high-dimensional data increases computational complexity and reduces prediction performance.

To overcome these problems, Principal Component Analysis (PCA)-based feature reduction techniques are combined with machine learning models such as Linear Regression, Support Vector Regression (SVR), Regression Tree, Random Forest, and K-Nearest Neighbors (KNN). PCA removes less important features and improves prediction efficiency while reducing execution time.

OBSERVATION TECHNIQUES

1. Historical Data Observation

Historical meteorological and operational data were continuously observed to identify electricity consumption patterns. The dataset included:

- Relative Humidity
- Dry Bulb Temperature
- Dew Point Temperature
- Ambient Temperature
- Wind Speed
- Atmospheric Pressure
- Solar Radiation
- Occupancy
- Lighting Load
- Heating and Cooling Load

These observations helped identify seasonal variations, daily energy usage trends, and peak demand periods.

2. Climate Zone Observation

Buildings located in four different climate zones were analyzed:

- Tabriz (Cold Climate)
- Tehran (Mild Climate)
- Yazd (Warm-Dry Climate)
- Bandar Abbas (Hot-Humid Climate)

Different climate zones showed different energy consumption behavior. Tabriz required more heating energy, while Bandar Abbas consumed higher cooling energy. This observation proved that climate conditions strongly affect prediction accuracy.

3. PCA-Based Feature Reduction Observation

Principal Component Analysis (PCA) was used to reduce dataset dimensions by transforming correlated variables into smaller uncorrelated variables called Principal Components (PCs).

The PCA loading plots identified:

- Strong influential features
- Weak influential features
- Correlated variables

Features with low loading values such as atmospheric pressure, month, wind direction, and relative humidity were removed to reduce computational complexity and improve efficiency.

4. Prediction Model Observation

Five prediction models were analyzed:

1. Linear Regression
2. SVR
3. Regression Tree
4. Random Forest
5. KNN

The dataset was divided into:

- 80% training data
- 20% testing data

The models were evaluated based on:

- Prediction accuracy
- Mean Square Error (MSE)

- R^2 Score
- Execution time

The observations showed that prediction performance depends strongly on climate patterns and historical data characteristics.

5. Energy Consumption Pattern Observation

Energy consumption patterns for heating, cooling, occupancy, and lighting loads were observed.

Key observations included:

- Cooling demand increased during daytime in hot climates.
- Heating demand increased during cold weather.
- Office occupancy significantly affected electricity usage.

These observations helped improve understanding of real-world building energy behavior.

6. Residual Error Observation

Residual plots were analyzed to study prediction errors.

Residual analysis helped identify:

- Model stability
- Error distribution
- Forecast reliability
- Overfitting issues

Low residual values indicated that predicted energy values were close to actual values.

ADVANTAGES

1. Dimensionality Reduction

PCA removes unnecessary features from the dataset.

Benefits include:

- Reduced memory usage
- Faster processing
- Lower computational complexity

2. Improved Prediction Accuracy

PCA-based prediction models improved forecasting accuracy by selecting only the most relevant features.

Random Forest and Regression Tree models achieved:

- $R^2 \approx 0.99$
- Very low MSE values

This indicates highly accurate prediction performance.

3. Reduced Execution Time

PCA preprocessing significantly reduced execution time.

Examples:

- SVR execution time reduced by approximately 50%
- Random Forest execution time reduced by approximately 25%

This makes the system suitable for real-time applications.

4. Noise Reduction

PCA removes noisy and weakly influential variables, improving:

- Forecast reliability
- Model stability
- Learning efficiency

5. Better Generalization Capability

The proposed models adapted effectively to different datasets and climate conditions, improving robustness and real-world deployment capability.

6. Climate-Specific Optimization

Different prediction models performed better for different climate zones.

Examples:

- Random Forest performed best for Tehran and Yazd.
- Regression Tree performed best for Tabriz and Bandar Abbas.

7. Scalability

The system can handle:

- Large datasets
- Multiple buildings

- Smart city energy systems

This makes it suitable for large-scale applications.

8. Smart Energy Management Support

The proposed approach supports:

- Smart grids
- HVAC optimization
- Energy-saving systems
- Carbon emission reduction

9. Better Resource Allocation

Accurate forecasting helps energy providers:

- Balance electricity supply
- Prevent overload conditions
- Reduce operational costs

10. Real-Time Monitoring Support

Reduced execution time enables integration into real-time monitoring and forecasting systems.

DISADVANTAGES

1. Information Loss

PCA removes some dataset features, which may lead to:

- Partial information loss
- Reduced interpretability
- Missing hidden relationships

2. Complex Mathematical Computation

PCA requires advanced matrix calculations and eigenvalue decomposition, making implementation difficult for beginners.

3. Dependency on Data Quality

Prediction accuracy depends heavily on dataset quality.

Problems such as:

- Missing values

- Noisy data
- Incorrect sensor readings

can reduce model performance.

4. Climate Dependency

Prediction models behave differently for different climate zones. A model performing well in one city may not perform well in another.

5. High Computational Requirements

Although PCA reduces dimensions, machine learning models still require:

- High RAM
- GPU acceleration
- Powerful processors

6. Difficult Hyperparameter Tuning

Machine learning models require careful tuning of parameters such as:

- Learning rate
- Number of trees
- Number of neighbors
- Kernel functions

Improper tuning decreases prediction accuracy.

7. Black-Box Nature

Models like Random Forest and SVR are difficult to interpret compared to traditional statistical methods.

8. Risk of Overfitting

Complex models may memorize training data instead of learning generalized patterns, reducing real-world performance.

9. Long Training Time

Large datasets require significant preprocessing and training time.

10. Hardware Cost

Real-time deployment requires:

- Dedicated servers

- Cloud infrastructure
- High-performance hardware

This increases implementation cost.

PERFORMANCE ANALYSIS

1. Linear Regression Performance

Linear Regression provided:

- Moderate accuracy
- Fast execution
- Low computational complexity

However, it struggled with nonlinear energy consumption patterns.

2. SVR Performance

SVR achieved better nonlinear prediction capability than Linear Regression but required higher execution time and computational power.

3. Regression Tree Performance

Regression Tree achieved:

- Very low MSE
- High R^2 values
- Fast execution

It performed especially well for Tabriz and Bandar Abbas datasets.

4. Random Forest Performance

Random Forest achieved the best overall performance with:

- Very high prediction accuracy
- Strong generalization capability
- Very low prediction error

The model achieved approximately $R^2 \approx 0.99$ for Tehran and Yazd datasets.

5. KNN Performance

KNN showed:

- Higher prediction error

- Lower scalability
- Reduced efficiency for large datasets

6. PCA Effectiveness Analysis

PCA preprocessing:

- Reduced execution time
- Improved computational efficiency
- Maintained prediction accuracy

Compared to random feature reduction, PCA-based reduction performed significantly better.

CONCLUSION

The PCA-based energy consumption prediction framework provides an effective solution for modern smart energy forecasting systems. Observation techniques such as historical data analysis, climate monitoring, PCA loading analysis, and residual analysis improved forecasting accuracy and reduced computational complexity.

Among all prediction methods, Random Forest and Regression Tree achieved the best performance depending on climate conditions. PCA-based preprocessing reduced execution time while maintaining high prediction accuracy.

Although the system introduces challenges such as computational complexity, climate dependency, and hardware requirements, it remains a highly effective solution for smart building energy management and future smart grid applications.